Online Examination System

Week 8 Practice - Question 3

Design: Student has many Attempts; an Examination has multiple Questions; Attempt stores answers and has InProgress/Submitted state.
Each question subtype implements its own evaluation, so adding a new type does not change submission logic.
Java solution:

```
import java.util.*;
abstract class Question {final String id;final int points;Question(String i,int
p){id=i;points=p;}abstract boolean correct(String answer);}
class MultipleChoiceQuestion extends Question {final String key;MultipleChoiceQuestion(String
i,int p,String k){super(i,p);key=k;}boolean correct(String a){return
key.equalsIgnoreCase(a);}}
class TrueFalseQuestion extends Question {final boolean key;TrueFalseQuestion(String i,int
p,boolean k){super(i,p);key=k;}boolean correct(String a){return
Boolean.toString(key).equalsIgnoreCase(a);}}
class ShortAnswerQuestion extends Question {final String key;ShortAnswerQuestion(String i,int
p,String k){super(i,p);key=k;}boolean correct(String a){return
a!=null&&key.equalsIgnoreCase(a.trim());}}
class Student {final String id;Student(String i){id=i;}}
class Examination {final String title;final List<Question> questions;Examination(String
t,List<Question> q){title=t;questions=new ArrayList<>(q);}}
class Attempt {enum Status{InProgress,Submitted}final Student student;final Examination
exam;private final Map<Question,String> answers=new HashMap<>();private Status
status=Status.InProgress;
Attempt(Student s,Examination e){student=s;exam=e;}void answer(Question q,String
a){if(status==Status.Submitted)throw new IllegalStateException("Cannot change answers for a
submitted examination.");if(!exam.questions.contains(q))throw new
IllegalArgumentException("Question not in exam");answers.put(q,a);}
int submit(){if(status==Status.Submitted)throw new IllegalStateException("Already
submitted");status=Status.Submitted;int score=0;for(Question
q:exam.questions)if(q.correct(answers.get(q)))score+=q.points;return score;}}
public class Main {public static void main(String[] args){Student s=new Student("Student
1");Question q1=new MultipleChoiceQuestion("Q1",5,"C"),q2=new
TrueFalseQuestion("Q2",5,false);Examination e=new Examination("Exam
A",Arrays.asList(q1,q2));Attempt a=new Attempt(s,e);
System.out.println("Exam A started by Student 1.");a.answer(q1,"C");System.out.println("Answer
recorded for Question 1.");a.answer(q2,"True");System.out.println("Answer recorded for
Question 2.");int score=a.submit();System.out.println("Exam A submitted by Student
1.");System.out.println("Result: Question 1: Correct (5 points), Question 2: Incorrect (0
points). Total score: "+score+"/10.");
try{a.answer(q1,"A");}catch(IllegalStateException ex){System.out.println(ex.getMessage());}}}
```

Expected score is 5/10 and the submitted attempt locks answers.